Novel Partitioning-Based Approach for Electromigration Assessment With Neural Networks

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***Abstract*—Due to continuing technology scaling, electromigration (EM) remains a prominent reliability concern in integrated circuit design. Traditional empirical methods often result in over-design in very large scale integration (VLSI) due to model inaccuracy. Recently, researchers have focused on analyzing EM susceptibility by tracking hydrostatic stress evolution in metal lines, governed by computationally expensive partial differential equations (PDEs). In this article, we propose a partitioning-based approach using neural networks to efficiently forecast the stress evolution along interconnect trees during the void nucleation and growth phases. This approach begins by decomposing the interconnect tree into subcomponents, providing computationally efficient analytical solutions for predicting stress evolution within each subtree. Subsequently, we employ a lightweight neural network to reassemble these components with their corresponding solutions to the original structure, ensuring accurate stress prediction. This divide-and-conquer strategy can accommodate various tree structures, with offshoots at arbitrary junctions, and holds substantial promise for using NN-based methods to solve mesh-free stress evolution on much larger interconnect trees than previously possible, with reduced computational overhead and heightened accuracy. The proposed approach eliminates the need for time discretization and grid meshing typically required in numerical methods. Numerical results confirm its advantages in accuracy and computational efficiency.**

***Index Terms*—Electromigration (EM), interconnect tree, machine learning (ML), partitioning.**

I. Introduction

ELECTROMIGRATION (EM) remains a major circuit reliability concern in advanced technology. This concern arises from the reduced service lifetime of interconnects under high current density, resulting in voids, a reduction of cross-sectional area, and an increase in line resistance. Thus, EM verification is crucial for chip signoff in very large scale integration (VLSI).

Traditional EM models have predominantly relied on empirical approaches, which often lack the accuracy needed for complex multiline structures. Based on the well-known Blech effect [1], a certain length-current density product can be used to identify EM-robust interconnects, called "immortal" wires. In additional, Black's equation [2] provides an empirical model for the mean-time-to-failure (MTF) of isolated metal lines, as a relationship between EM failure time, current density, and temperature. However, applying these methods to multiline metal structures implicitly assumes that individual lines may be modelled as independent isolated lines, which is not true and leads to inaccurate reliability evaluation when applied to on-chip circuits, such as power grids [3]. As an alternative physical model, and building on previous work in [4], [5], and [6], EM assessment based on Korhonen's equation [7] effectively captures the physical kinetics. This equation describes the relationship between the current density-driven electron wind force and the diffusion gradient-driven back stress force along a wire, formulated as partial differential equations (PDEs). According to this model, tree-level analysis for EM-induced stress calculation has been proposed to consider the material flow between adjacent segments confined by diffusion barriers, rather than solely focusing on individual wires [8], [9], [10]. Solving the PDEs-governed physical model remains a challenge, especially as interconnect tree sizes grow. Numerical methods, such as the finite difference method (FDM) and finite element method (FEM), are typically employed for stress analysis, necessitating time discretization and mesh refinement [11], [12]. Analytical methods, on the other hand, yield closed-form solutions for hydrostatic stress via Laplace transformations [13], [14], [15], offering computational acceleration for specific interconnect structures. Recently, machine learning (ML) has emerged as a promising tool for modeling the evolution of EM-induced stress governed by complex PDEs.

To address the scalability issues in large interconnect trees, we partition the EM-induced stress problem and incorporate subproblems. Partitioning techniques have traditionally been used in VLSI to accelerate the solution of large-scale problems by leveraging circuit locality. A large circuit can be divided into smaller subcircuits connected at a few global boundaries. This divide-and-conquer strategy reduces computational complexity by manipulating the entire system into hierarchical tree

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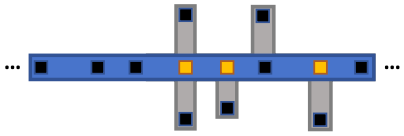


Fig. 1. Example of an interconnect tree consisting of a long stripe (blue) with occasional offshoots (gray) from certain junctions. Black blocks denote junctions in the interconnect tree; yellow blocks indicate selectable joints in the proposed approach.

structures after optimization, scaling, relaxation, partitioning, and assignment [16], [17].

This study focuses on EM assessment in power grids, where metal lines mostly carry unidirectional currents and are therefore susceptible to EM failure. Fig. 1 shows an example of a typical interconnect tree in modern power grids, where the tree consists of a long stripe (blue) with possibly occasional offshoots (gray) from certain junctions. We introduce a novel learning approach to calculate stress evolution along such interconnect trees with offshoots. Our main contributions includes as follows.

1) We propose, for the first time, a fast partitioning-based approach to predict mesh-free stress evolution along interconnect trees during both the void nucleation (VN) and growth phases. The method is based on a neural network and requires no prior knowledge of the stress solution. Designed to align with EM analysis, the proposed framework considers physical parameters like varying segment width and random diffusivity, as well as operating conditions like time-dependent temperature and dynamic current densities caused by EM-induced metal line degradation.
2) We introduce a novel divide-and-conquer strategy that enables multiple partitioning approaches for the original interconnect tree and derives computationally efficient stress solutions for the subtrees. This approach results in significant computational savings, with additional time reduction achieved through parallelization.
3) We develop a learning approach to combine the stress solutions of the subtrees, leveraging the locality of EM stress evolution, which reduces the number of constraint conditions in the learning task compared with the existing methods. The proposed approach is extended to substantially improve the convergence speed by incorporating random Fourier feature (RFF) embeddings.

The remainder of the article is structured as follows. Section II provides a brief overview of the related works, EM physics, and stress modeling, along with an introduction to spectral bias and RFFs. Section III outlines the partitioning method and derives the brick-joint solution. Section IV introduces the learning framework and extends it to solve stress evolution under dynamic current densities. Section V presents the results of the proposed method and compares its performance against competing methods. Finally, Section VI concludes this article and this article ends with an Appendix that includes additional technical details.

II. Preliminary

We will review several ML-based approaches for EM assessment, along with key fundamentals and preliminaries for modeling EM physics and establishing the neural network-based framework.

A. Related Work

ML has shown effectiveness in modeling various physical systems, including capturing the evolution of EM-induced hydrostatic stress [18], [19], [20], [21], [22], [23]. Recently, physics-informed neural networks (PINN) have been introduced to model physical dynamics by encoding constraints via PDEs [24], providing a novel approach to solving Korhonen's equation. Studies in [19], [20], and [23] have shown promising results in ML-aided EM analysis by employing a coordinate-based neural representation to achieve stress evolution on interconnect trees with multiple segments.

B. EM Physics and Stress Modeling

EM occurs within metal wires carrying significant current densities, where metal atoms are bombarded by the electron flow. This resulting atomic flow increases compressive stress at the anode and tensile stress at the cathode, presenting an opposing driving force that impedes EM progress [1]. In this section, we first introduce the physics-based modeling describing the stress evolution and then present the concept of stress flow [14], which is employed to construct the analytical solution of stress evolution along linear interconnect trees during the VN phase.

1) Physics-Based Modeling: The evolution of EM-induced stress can be characterized by the 1-D Korhonen's equation [7], which combines the atomic flux equation tracking the atomic mass transport, and the continuity equation enforcing the mass balance. The atomic flux describing the migration of metal atoms within the confined line can be expressed as

$$\phi(x) = \frac{DC}{k_b T}\left(\Omega\frac{\partial \sigma}{\partial x} + q^*\rho j\right) \tag{1}$$

where j and x represent the current density and location, respectively. The effective atomic diffusivity $D = D_0 \exp(-E_a/k_bT)$ depends on the activation energy E_a, the self-diffusion coefficient D_0, the Boltzmann constant k_b and the temperature T. The variables C, q^*, Ω, and ρ denote the atomic concentration, effective charge, atomic lattice volume, and metal resistivity, respectively. Furthermore, the continuity equation, which accounts for the interplay between stress evolution over time and material transport, yields the diffusion equation

$$\frac{\partial \sigma(x,t)}{\partial t} = \frac{\partial}{\partial x}\left[\kappa\left(\frac{\partial \sigma(x,t)}{\partial x} + G\right)\right] \tag{2}$$

where $\kappa = DB\Omega/k_bT$ and $G = q^*\rho j/\Omega$. The notation B is the effective bulk modulus associated with the line geometry, especially width, aspect ratio and grain morphology [25], [26].

Fig. 2 illustrates interconnect trees featuring degree-1 to 4 junctions, where the degree indicates the number of lines

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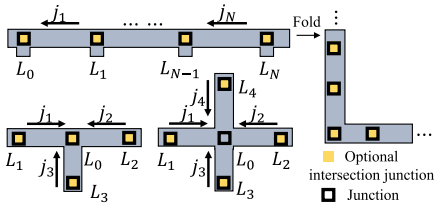


Fig. 2. General straight multisegment, T-shaped, and cross-shaped interconnect trees. A straight multisegment brick can be transformed by folding the segments at the junctions. Notations L_k and j_i represent the location of the kth node and the current density loaded within the ith segment. L_0 is equal to zero. The hollow square represents the junction of the interconnect tree. The yellow square indicates an optional intersection junction within the partitions.

connecting to the junction (hollow squares). Boundary conditions (BCs) for the junctions should be established in the EM model. A preliminary step is to enforce the balance between the outgoing and incoming atomic flux at junctions with a degree greater than one (called interior junction), which we refer to as flux balance

$$\sum_{i\in S_r} \pm W_i \cdot \kappa_i \left(\frac{\partial \sigma_i(x,t)}{\partial x}\Big|_{x=x_r} + G_i \right) = 0 \tag{3}$$

where the atomic flux term is positive in the incoming direction. W_i is the branch width of the ith segment, S_r is the index set of all the segments connected to the rth interior junction and (3) demonstrates the net atomic flux entering the rth interior junction is zero. Moreover, the stress continuity condition is enforced at the interior junctions, as the stress evolution must remain continuous along the tree

$$\sigma_{S_r[1]}(x_r,t) = \cdots = \sigma_{S_r[m]}(x_r,t). \tag{4}$$

BC at degree-1 junctions (called terminal junction) and the initial condition for interconnect trees can be categorized into two cases: 1) VN phase when no EM-induced voiding occurs in metal lines and 2) VG phase when voids are created and develop over time.

1) *Void Nucleation:* Before void occurs, the atomic flux ϕ at the diffusion barrier equals zero. BC at terminals follows:

$$\kappa_t \left(\frac{\partial \sigma_t^n(x,t)}{\partial x}\Big|_{x=x_t} + G_t \right) = 0 \tag{5}$$

where σ^n shows the stress evolution during the VN phase as indicated by the superscript n, and x_t denotes the location of the terminal. For each segment, the initial condition can be written as

$$\sigma_i^n(x,0) = \sigma_T \tag{6}$$

where σ_T denotes the residual stress at the initial time, which is uniform as a result of stress migration occurring immediately after chip manufacturing.

2) *Void Growth:* once a void occurs, typically after the stress evolution reaches a critical level defined as σ_{crit} (this moment is called the VN time t_{nuc}), the normal component of stress at the void surface is constrained to zero due to the absence of confinement at the surface. However, the stress nearby cannot change immediately and will gradually decrease over time. The infinitesimally small distance between the zero stress and the remaining stress is referred to as the effective thickness of the void interface, denoted as δ_s. In [27], this value is set to $\delta_s = 1$ nm. The tree is subsequently divided into several subtrees based on the void formation location. Typically, the voids will be situated at the terminal junctions of the decomposed trees [10], [11], [28], as it is supported by the observation that flux divergence is higher around junctions and especially around vias. Assuming a linear stress distribution across the void interface, the stress gradient follows $d\sigma_v^g(x_v,t)/dx = \pm\sigma_v^g(x_v,t)/\delta_s$ where g denotes the VG phase and x_v is the location of the void surface. For simplicity, we rewrite the BC at the voided terminal as [15]

$$\sigma_v^g(x_v,t) = 0 \tag{7}$$

where σ^g is the stress evolution during the VG phase. The initial stress along the interconnect tree, excluding the void surface, remains the same as the stress evolution at t_{nuc}. Therefore, we have

$$\sigma_i^g(x,0) = \sigma_i^n(x,t_{nuc}). \tag{8}$$

2) Stress Flows: For a straight multisegment interconnect, stress flow components are introduced in [14] to demonstrate the contribution of sources located at terminals and intersections of the interconnect trees to the stress at location x. These stress flows are the fundamental terms to build up the series in the stress solution, and the corresponding traveled distance, defined as $X_i^k(x)$, follows:

$$X_i^0(x) = \begin{cases} iL_N + x, & i \text{ is even} \\ (i+1)L_N - x, & i \text{ is odd} \end{cases}$$
$$X_i^N(x) = \begin{cases} (i+1)L_N + x, & i \text{ is even} \\ iL_N - x, & i \text{ is odd} \end{cases}$$
$$X_i^k(x) = \begin{cases} L_k + iL_N - x, & i \text{ is even, to the left} \\ L_k + (i-1)L_N + x, & i \text{ is odd, to the left} \\ -L_k + iL_N + x, & i \text{ is even, to the right} \\ -L_k + (i+1)L_N - x, & i \text{ is odd, to the right.} \end{cases} \tag{9}$$

where k represents the distance from the kth junction to x and satisfies $k = 1, \ldots, N-1$. The stress flow reflects at the boundaries and i represents the ith reflection. Left and right flows are created for the source at interior junctions and travel sideways from the source. The 0th reflection distance from the source of interior junctions called fundamental traveled distances, are combined from the left and right sides at $x = L_k$ into the distance $X_0^k(x) = |L_k - x|$. The stress flow decays exponentially with distance traveled and is obtained by [13]

$$g(x,t) = 2\sqrt{\frac{\kappa t}{\pi}} e^{-\frac{x^2}{4\kappa t}} - x \times \texttt{erfc}\left\{ \frac{x}{2\sqrt{\kappa t}} \right\} \tag{10}$$

demonstrating the relationship between the stress flow, traveled distance, and time. The notation `erfc` represents the complementary error function. Finally, the stress flow under the ith reflection is defined as $\pm G_k g(X_i^k(x),t)$ for $k = 0, N$, and $(G_{k+1} - G_k) g(X_i^{k,l/r}(x),t)/2$ for $0 < k < N$ where $X_i^{k,l}(x)$ and $X_i^{k,r}(x)$ are the left and right traveled distance in $X_i^k(x)$.

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C. Spectral Bias Analysis Through Neural Tangent Kernel and Random Fourier Feature Embedding

The recently developed neural tangent kernel (NTK) theory [29], [30] bridges the spectral bias analysis with the training behavior of fully-connected neural networks [31]. Given a dataset consisting of input $X_{\text{train}} = [\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n] \in \mathbb{R}^{d\times n}$ and expected output $Y_{\text{train}} = [y_1, \ldots, y_n]$, and considering a network $f(\cdot)$ with unknown weights θ trained by minimizing the loss function $\text{MSE} = ||f(\theta; X_{\text{train}}) - Y_{\text{train}})||_2^2$, the entries of NTK, the dot-product kernel, are given by [29] and [32]

$$K_{ij} = K(\boldsymbol{x}_i, \boldsymbol{x}_j) = \left\langle \frac{df(\theta;\boldsymbol{x}_i)}{d\theta}, \frac{df(\theta;\boldsymbol{x}_j)}{d\theta} \right\rangle \tag{11}$$

demonstrating the gradient descent dynamics under an infinitely small learning rate. The convergence rate of network training is shown to be determined by the eigenvalues of the kernel [30], [31]. It is suggested in [33] that a conventional fully-connected neural network is incapable of learning high-frequency tasks, where the eigenvalues of the kernel decrease sharply and monotonically with increasing frequency of the eigenvectors, a phenomenon termed "spectral bias."

To overcome this problem, [31] and [34] introduce RFF embeddings into the neural network framework, defined as

$$\gamma(\boldsymbol{B}, \boldsymbol{x}) = [\cos(\boldsymbol{Bx}), \sin(\boldsymbol{Bx})]^T \tag{12}$$

where $\boldsymbol{B} = [\boldsymbol{b}_1, \boldsymbol{b}_2, \ldots, \boldsymbol{b}_m]^T \in \mathbb{R}^{m\times d}$ is sampled from a Gaussian distribution $\mathcal{N}(0, \nu^2)$ and ν is a fixed hyperparameter. The inputs $\boldsymbol{x}$ are transformed through the Fourier mapping $\gamma(\cdot)$, followed by a fully connected neural network. It has been demonstrated in [31] that the kernel of a two-layer linear and bias-free neural network with RFF embeddings follows $K_\gamma(x, x') = \cos(b(x - x'))$, featuring nonzero eigenvalues $\lambda = (1 \pm \sin b/b)/2$ by setting $\boldsymbol{B} = b \in \mathbb{R}$ when $d = m = 1$. The corresponding eigenfunction is $g(x) = C_1 \cos(bx) + C_2 \sin(bx)$, where C_1 and C_2 are constants. The finding illustrates that training behavior with larger ν would yield higher-frequency eigenfunctions and smaller eigenvalue gaps, potentially mitigating the spectral bias issue and accelerating convergence speed.

III. Novel Partitioning-Based Approach

We start with a general interconnect tree with the goal of selecting certain junction nodes to be removed, which disconnects the tree, and refer to these nodes as "*joints*." This approach draws inspiration from node tearing technology [35].

A. Analytical Stress Solutions for Individual Bricks

The schematic of each brick, introducing the notations L_k and j_i, and the positive current direction, is illustrated in Fig. 2. We denote the initial time in each EM phase by t_0, which is set to 0 during the VN phase and t_{nuc} during the VG phase. The notation $\mathcal{T} = t - t_0$ is then introduced to represent the evolution time, where t denotes the observation time instance.

1) Straight Multisegment Brick: During the VN phase, [14] demonstrates that substituting (9) and (10) into

$$g_{i,k}^n(x,t) \triangleq \begin{cases} g\left(X_i^k(x), t\right), k = 0, N \\ \frac{g\left(X_i^{k,l}(x),t\right) + g\left(X_i^{k,r}(x),t\right)}{2}, k = 1, \ldots, N-1 \end{cases} \tag{13}$$

where, as before, $X_i^{k,l}(x)$ and $X_i^{k,r}(x)$ are the left and right traveled distances of $X_i^k(x)$ from the kth junction during the ith reflection, the analytical stress evolution on a straight multisegment brick follows:

$$\sigma_{2,j}^n(x,t) = \sum_{i=0}^{+\infty}\sum_{k=0}^{N} A_k g_{i,k}^n(x, \mathcal{T}), L_{j-1} \le x \le L_j \tag{14}$$

where the subscript $2, j$ represents the jth segment of a straight N-segment brick (in which the maximum degree of a junction is "2"), while A_k equals G_1 for $k = 0$, $-G_N$ for $k = N$ and $(G_{k+1} - G_k)$ for $k = 1, \ldots, N-1$.

During the VG phase, for $t > t_{nuc}$, since the analytical solution has already been published in [15], we summarize the solution below. Let $f_k(x) = \sigma_{2,k}^n(x, t_{nuc})$ where $x \in [L_{k-1}, L_k]$. The influence of initial stress along the kth segment can be computed as [15]

$$b_k(m, x, t) = \frac{1}{L_N} e^{-M_b} \int_{L_{k-1}}^{L_k} f_k(u) \sin(z_m(u - x))du \tag{15}$$

where $z_m = (m - 0.5)\pi/L_N$ is the mth zeros obtained from the Residue Theorem and we have $M_b = z_m^2 \kappa_k t$. Consistent with a decay phenomenon, (15) is close to zero as time increases to infinity. The overall impact of initial stress on stress evolution along the jth segment at $t > t_{nuc}$ follows:

$$\mathcal{B}_{2,j}^g(x,t) = \sum_{m=0}^{+\infty}(-1)^m \left\{ \sum_{k=1}^{N} b_k\left(m, x^-, \mathcal{T}\right) - \sum_{k=1}^{j-1} b_k\left(m, -x^-, \mathcal{T}\right) + \sum_{k=j}^{n} b_k\left(m, x^+, \mathcal{T}\right) \right\} \tag{16}$$

where $x^+ = L_N + x$ and $x^- = L_N - x$. Then we establish

$$g_{i,k}^g(x,t) \triangleq \begin{cases} (-1)^i g_0, k = 0 \\ \frac{g_0 - g\left(\eta_{N,k}^{j-}(i,x),t\right) + g\left(\xi_{N,k}^{j+}(i,x),t\right)}{2\times(-1)^i}, 1 \le k < j \\ \frac{g_0 + g\left(\eta_{N,k}^{j-}(i,x),t\right) - g\left(\xi_{N,k}^{j+}(i,x),t\right)}{2\times(-1)^i}, j \le k < N \end{cases} \tag{17}$$

where $g_0 = -g(\xi_{N,k}^{j-}(i,x), t) + g(\eta_{N,k}^{j+}(i,x), t)$ and the detailed notations $\eta_{N,k}^{j-}$, $\eta_{N,k}^{j+}$, $\xi_{N,k}^{j-}$, $\xi_{N,k}^{j+}$ can be found in [15], representing the traveled distances. Stress evolution along the jth segment of a straight interconnect tree satisfies [15]

$$\begin{aligned} \hat{\sigma}_{2,j}^g(x,t) &= \sigma_{2,j}^g(x,t) + \mathcal{B}_{2,j}^g(x,t) \\ &= \sum_{i=0}^{+\infty}\sum_{k=0}^{N} A_k g_{i,k}^g(x, \mathcal{T}) + \mathcal{B}_{2,j}^g(x,t) \end{aligned} \tag{18}$$

where $A_N = 0$ and the void formation is assumed at the right terminal ($x = L_N$).

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Algorithm 1 Vectors Used for Stress Evolution Assessment

Input: Max iteration number i_{max}, index of segment j, index of junction where void occurs v (v = NA during void nucleation phase), number of segments M, length of the i-th segment L_i, a set of indices of segments C;

Output: Vectors $\boldsymbol{\mathcal{H}}^C$, $\boldsymbol{\mathcal{H}}^*$, $\boldsymbol{h}^C$, $\boldsymbol{h}^*$, $\mathbf{K}^{i,M}$, $\mathbf{L}^{i,M}$.

Initialize empty $\boldsymbol{\mathcal{H}}^C = [\,]$, $\boldsymbol{h}^C = [\,]$.

Initialize $\mathbf{L}^{0,M} = [0]$, $\mathbf{L}^{1,M} = [2\sum_{i=1}^{M} L_i]$, $\mathbf{K}^{0,M} = [1]$ and $\boldsymbol{\mathcal{H}}^* = [\sum_{\substack{i=1,\\ i\neq j}}^{M} L_i, -\sum_{\substack{i=1,\\ i\neq j}}^{M} L_i]$.

if $v ==$ **NA then**
  $\mathbf{K}^{1,M} = [1]$, $\boldsymbol{h}^* = [M, -\lfloor M/2 \rfloor]$,
else
  $\mathbf{K}^{1,M} = [-1]$, $\boldsymbol{h}^* = [M, \lfloor M/2 \rfloor]$,
end if
for $c = 1 \rightarrow M$ **do**
  if $c \notin C$ **then**
    $\boldsymbol{\mathcal{H}}^C = \mathcal{F}((\boldsymbol{\mathcal{H}}^C)^T + [c, -c])$,
    if $c == v$ **then**
      $\boldsymbol{h}^C = \mathcal{F}((\boldsymbol{h}^C)^T \cdot [1, -1])$,
    else
      $\boldsymbol{h}^C = \mathcal{F}((\boldsymbol{h}^C)^T \cdot [1, 1])$,
    end if
  end if
  if $c \neq j$ **then**
    $\boldsymbol{\mathcal{H}}^* = \boldsymbol{\mathcal{H}}^* \| -2L_c + \sum_{\substack{i=1,\\ i\neq j}}^{M} L_i$,
    if $c == v$ **then**
      $\boldsymbol{h}^* = \boldsymbol{h}^* \| -\lfloor M/2 \rfloor$,
    else
      $\boldsymbol{h}^* = \boldsymbol{h}^* \| \lfloor M/2 \rfloor$,
    end if
  end if
  $\mathbf{L}^{1,M} = \mathbf{L}^{1,M} \| 2L_c \| 2\sum_{\substack{i=1,\\ i\neq n}}^{M} L_i$,
  if $v ==$ NA **then**
    $\mathbf{K}^{1,M} = \mathbf{K}^{1,M} \| \frac{2-M}{M} \| \frac{M-2}{M}$,
  else if $c == v$ **then**
    $\mathbf{K}^{1,M} = \mathbf{K}^{1,M} \| \frac{M-2}{M} \| \frac{M-2}{M}$,
  else
    $\mathbf{K}^{1,M} = \mathbf{K}^{1,M} \| \frac{2-M}{M} \| \frac{2-M}{M}$,
  end if
end for
for $i = 2 \rightarrow i_{\max}$ **do**
  $\mathbf{L}^{i,M} = \mathcal{F}((\mathbf{L}^{1,M})^{\mathrm{T}} + \mathbf{L}^{i-1,M})$,
  $\mathbf{K}^{i,M} = \mathcal{F}((\mathbf{K}^{1,M})^{\mathrm{T}} \cdot \mathbf{K}^{i-1,M})$.
end for

2) T-Shaped and Cross-Shaped Bricks: In the following, the Laplace transform is used to derive the analytical stress solutions for bricks with a degree-M junction, where $M = 3$ for the T-shaped brick and $M = 4$ for the cross-shaped brick. A detailed derivation process for stress evolution along a T-shaped brick during the VN phase is given in Appendix A. In Algorithm 1 of Appendix A, we provide a set of vectors, including $\mathbf{L}^{i,M}$, $\boldsymbol{\mathcal{H}}^C$, and $\boldsymbol{\mathcal{H}}^*$, where i is a non-negative integer related to the number of stress flow reflections. We define $\boldsymbol{\mathcal{H}}^C \triangleq \boldsymbol{\mathcal{H}}(C)$ as a transformation that maps C, a set of indices of segments, to a vector used for calculating stress evolution. These vectors are then employed to obtain the traveled distance

$$\begin{aligned}
\xi^{j+}_{p,q}(i,x) &= \mathbf{L}^{i,M}_p - \boldsymbol{\mathcal{H}}^{\{j\}}_q - L_\Sigma - L_j + x \\
\xi^{j-}_{p,q}(i,x) &= \mathbf{L}^{i,M}_p - \boldsymbol{\mathcal{H}}^{\{j\}}_q - L_\Sigma + L_j - x \\
\eta^{j+}_{p,q,k}(i,x) &= \begin{cases} \mathbf{L}^{i,M}_p - \boldsymbol{\mathcal{H}}^{\{j,k\}}_q - L_\Sigma - L_j + x, & j \neq k \\ \mathbf{L}^{i,M}_p + \boldsymbol{\mathcal{H}}^{*}_q - L_\Sigma + x, & j = k \end{cases} \\
\eta^{j-}_{p,q,k}(i,x) &= \begin{cases} \mathbf{L}^{i,M}_p - \boldsymbol{\mathcal{H}}^{\{j,k\}}_q - L_\Sigma + L_j - x, & j \neq k \\ \mathbf{L}^{i,M}_p - \boldsymbol{\mathcal{H}}^{*}_q - L_\Sigma - x, & j = k \end{cases}
\end{aligned} \tag{19}$$

where $x \in [L_0, L_j]$, $L_\Sigma = \sum_{j=1}^{M} L_j$, $j \in [1, M]$ and $k \in [0, M]$. The subscripts p and q are vector indices that traverse the entire vectors. At location x on the jth segment of the brick, we employ (20) to calculate the stress flow component from the source of the terminal junction on the kth segment

$$g^{n/g}_{i,j,k}(x,t,M,\boldsymbol{n}) = \begin{cases} \frac{\sum_{p,q} \mathbf{K}^{i,M}_p \boldsymbol{h}^{\{j\}}_q \left(g\left(\xi^{j+}_{p,q}(i,x),t\right) + \boldsymbol{n}_1 g\left(\xi^{j-}_{p,q}(i,x),t\right)\right)}{M}, & k = 0 \\ \frac{\sum_{p,q} \mathbf{K}^{i,M}_p \boldsymbol{h}^{*}_q \left(\boldsymbol{n}_2 g(\eta^{j+}_{p,q,j}(i,x),t) + g(\eta^{j-}_{p,q,j}(i,x),t)\right)}{M}, & k = j \\ \frac{2\sum_{p,q} \mathbf{K}^{i,M}_p \boldsymbol{h}^{\{j,k\}}_q \left(g\left(\eta^{j+}_{p,q,k}(i,x),t\right) + \boldsymbol{n}_3 g\left(\eta^{j-}_{p,q,k}(i,x),t\right)\right)}{M}, & k \neq 0, j \end{cases} \tag{20}$$

where $k = 0$ corresponds to the degree-M interior junction. The vectors $\mathbf{K}^{i,M}$, $\boldsymbol{h}^*$, and $\boldsymbol{h}^C$ can be calculated through Algorithm 1 of Appendix A where $\boldsymbol{h}^C \triangleq \boldsymbol{h}(C)$ represents a transformation from the set C to a vector, and $\boldsymbol{n} = [n_1, n_2, n_3]$ varies from different EM phases. The stress evolution on the jth segment of the brick with a degree-M junction follows:

$$\sigma^{n/g}_{M,j}(x,t) = \sum_{i=0}^{+\infty} \sum_{k=0}^{M} A_k g^{n/g}_{i,j,k}(x, \mathcal{T}, M, \boldsymbol{n}), L_0 \leq x \leq L_j \tag{21}$$

where $A_k = \sum_{i=1}^{M} G_i$ for $k = 0$, and $A_k = -G_k$ for $k = 1, \ldots, M$, and $\boldsymbol{n} = [1, 1, 1]$ when estimating $\sigma^{n}_{M,j}(x,t)$ during the VN phase (cf. Appendix A).

During the VG phase, we assume the void is created at the terminal junction on the vth segment. Let $f_k(x) = \sigma^{n}_{M,k}(x, t_{nuc})$. By utilizing the Residue Theorem, the initial unit function describing the influence of the initial stress along the kth segment on the stress along the jth segment yields

$$b_{j,k,M}(m,x,t) = \begin{cases} \frac{4e^{-M_b}\beta_j(z_m x)\prod_{\substack{i=1,\\ i\neq k,j}}^{M} \beta_i(z_m L_i) \int_{L_0}^{L_k} f_k(u)\beta_k(z_m(L_k-u))du}{Z}, & j \neq k \\ \frac{4e^{-M_b}\alpha_j(z_m x)\int_{L_0}^{L_k} f_k(u)\beta_k(z_m(L_k-u))du}{Z}, & j = k \end{cases} \tag{22}$$

where $\alpha_k(\cdot)$ and $\beta_k(\cdot)$ are $\cos(\cdot)$ and $\sin(\cdot)$ for $k = v$, and $\sin(\cdot)$ and $\cos(\cdot)$ for $k \neq v$. The notation z_m is the mth zeros of $p_M(z) = M\cos(L_\Sigma z) + (M-2)(\sum_{\substack{i=1,\\ i\neq v}}^{M} \cos((L_\Sigma - 2L_i)z) - \cos((L_\Sigma - 2L_v)z))$. Due to the periodicity of $p_M(z)$, the number of zeros is infinite. We have $Z = -dp(z)/dz|_{z=z_m}$ and $M_b = z_m^2 \kappa_k t$. The overall impact on stress evolution along the jth segment takes the form

$$\mathcal{B}^{g}_{M,j}(x,t) = \sum_{m=0}^{+\infty} \left\{ \sum_{k=1}^{M} \left[n_k b_{j,j,M}(m, x_k, \mathcal{T}) + b_{j,j,M}\left(m, x^+, \mathcal{T}\right) \right] \right.$$

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$$+ 2^{M-1} \sum_{\substack{k=1,\\ k\neq j}}^{M} b_{j,k,M}\left(m, L_j - x, \mathcal{T}\right) \Bigg\} \tag{23}$$

where $x^+ = x + \sum_{\substack{i=1,\\ i\neq j}}^{M} L_i$. Notations x_k and n_k are $x - \sum_{\substack{i=1,\\ i\neq j}}^{M} L_i$ and $2 - M$ for $k = v$, and $x^+ - 2L_k$ and $M - 2$ for $k \neq v$. Finally, during the VG phase, the stress evolution along the jth segment can be obtained

$$\hat{\sigma}^g_{M,j}(x,t) = \sigma^g_{M,j}(x,t) + \mathcal{B}^g_{M,j}(x,t) \tag{24}$$

where in (21) $\boldsymbol{n} = [-1, 0, -1]$ for $j = v$ and $\boldsymbol{n} = [1, -1, 1]$ for $j \neq v$.

B. Modified Analytical Stress Solutions for Bricks

By utilizing a linear transformation, EM-induced stress evolution, denoted by $\hat{\sigma}^{n/g}_{M,j}(x,t)$, can be converted to the sum of two stress components satisfying Korhonen's equation.

According to (14), (18), (21) and (24), we summarize the first stress component on individual N-segment bricks in which the maximum degree is M ($M = 2, 3, 4$)

$$\sigma^{n/g}_{M,j}(x,t) = \sum_{i,k} \hat{A}_k g^{n/g}_{i,k}(x, t - t_0), t > t_0 \tag{25}$$

where $i \in [0, \infty)$ and $k \in [0, N]$ are integers, $g^{n/g}_{i,k}(\cdot)$ follows (13), (17), and (20) for different M. Let S_k be a set of indices of the segments connected to the kth junction. The BC of (25) regarding atomic flux at junctions is governed by $\sum_{i\in S_k} \pm \partial \sigma^{n/g}_{M,i}(x,t)/\partial x|_{x=L_k} = \hat{A}_k$, where $\hat{A}_k$ represents the sum of EM driving forces flowing toward the kth junction. The spatial derivative of stress at junctions, expressed as $\partial\sigma(x,t)/\partial x|_{x=L_k}$, is defined as the stress gradient, the sum of which can be computed by the flux balance condition (3). We refer to the stress gradients in the same brick as local stress gradients. For bricks without intersection junctions, $\hat{A}_k$ is consistent with A_k in Section III-A. As indicated in [15], when $\hat{A}_k$ is time-dependent, (25) turns to

$$\sigma^{n/g}_{M,j}(x,t) = \sum_{i,k} \left\{ \hat{A}_k(t_0) g_{i,k}(x, t - t_0) + \hat{A}'_k(t) * g_{i,k}(x, t - t_0) \right\} \tag{26}$$

where $t > t_0$, and $\hat{A}'_k(t)$ represents the time derivative of $\hat{A}_k(t)$ and $*$ is the temporal convolution $a(t)*b(t-t_0) = \int_{t_0}^{t} a(\tau) b(t - t_0 - \tau) d\tau$.

1) Combination of Stress on Bricks: Assuming the lth junction is the intersection junction, the sum of local stress gradients toward the junction cannot be directly calculated due to the unknown stress and atomic flux from the neighboring brick. Compared with our previous work [21] that proposed a method to combine stress on intersected single metal lines, we develop a novel approach to enable the combination of stress on versatile interconnect structures by introducing the well-tailored $\hat{A}_k(t)$

$$\hat{A}_k(t) = \begin{cases} A_k, \ t = 0, k \neq l \\ m_c \frac{N_A W_n \kappa_n + A_k W \kappa}{m_n W_n \sqrt{\kappa_n \kappa} + m_c W \kappa}, \ t = 0, k = l \\ \hat{A}_k(0) + \int_0^t \hat{A}'_k(t) dt, \ t > 0 \end{cases} \tag{27}$$

where A_k is provided in Section III-A, the second term is motivated by [21] and the third term presents a method to compute $\hat{A}_k(t)$ when t is positive. Notations m_c and m_n are the number of segments connected with the intersection junction in the current and neighboring bricks. The notations W, κ and W_n, κ_n are the width, diffusivity of the metal branch of the current and neighboring brick, respectively. N_A equals the sum of EM driving forces toward the intersection junction in the neighboring brick and for the current brick, the temporal derivative term $\hat{A}'_k(t)$ is zero when $k \neq l$ and the current density within each brick is constant.

2) Transition Between EM Phases: Once an interconnect tree turns into the VG phase, not all of its bricks undergo this phase simultaneously. Bricks that remain in the VN phase will exhibit the stress evolution at t_{nuc} as the initial condition. Since the stress evolution can be decomposed into two components, the first stress component, with the initial condition $\sigma^g_{M,j}(x, t_{nuc}) = 0$ at the initial time $t_0 = t_{nuc}$ can be computed by (26). Furthermore, to estimate the second stress component $\mathcal{B}^n_{M,j}(x,t)$, the term $\hat{A}_k(t)$ in the VN phase becomes $\hat{A}^-_k(t) = \hat{A}_k(t)(1 - u(t - t_{nuc}))$ where $u(t)$ is the continuous-time unit step signal, used to set G_k to zero after t_{nuc}. In this EM phase, $\hat{A}^{-'}_l(t)$ is a determined function learned from the simulation in the VN phase. Then the derivative term yields

$$\frac{d\hat{A}^-_k(t)}{dt} = \hat{A}^{-'}_k(t) - \hat{A}_k(t_{nuc})\delta(t - t_{nuc}), t > 0 \tag{28}$$

where $\hat{A}^{-'}_k(t) = \hat{A}'_k(t)(1 - u(t - t_{nuc}))$. Substituting $t_0 = 0$ into (26), and replacing $\hat{A}_k(t_0)$ by $\hat{A}^-_k(t)$ applied at $t = 0$ and $\hat{A}'_k(t)$ by the right-hand side of (28), the stress decay $\mathcal{B}^n_{M,j}(x,t)$ can be obtained. This result is then added to the first stress component, achieving stress evolution during the VN phase considering the initial stress at t_{nuc}

$$\hat{\sigma}^n_{M,j}(x,t) = \sum_{k,i} \{ \hat{A}^-_k(0) g^n_{i,k}(x,t) + \int_0^t \hat{A}^{-'}_k(\tau) g^n_{i,k}(x, t - \tau) d\tau + \int_{t_{nuc}}^t \hat{A}'_k(\tau) g^n_{i,k}(x, t - t_{nuc} - \tau) d\tau \}, t > t_{nuc} \tag{29}$$

where $\hat{A}'_l(t)$ is unknown for $t > t_{nuc}$. When the interconnect tree is in the VN phase, stress solution $\hat{\sigma}^n_{M,j}(x,t)$ can be obtained by substituting $t_0 = 0$ into (26) where $\hat{A}'_l(t)$ remains unknown.

3) Stress Solution Under Dynamic Current Density and Temperature: When the current density within each segment is not constant, the dynamic EM driving force can be represented by $G_k(t)$ and used to calculate the time-dependent A_k. By substituting A_k into (27), the influence of dynamic current density can be considered where $\hat{A}'_k(t)$ ($k \neq l$) becomes nonzero. The impact of time-dependent temperature on stress can be addressed using the method in [15].

C. Bricks and Joints Building

1) Brick Solution: We first employ $\hat{A}^{+'}_l(t)$ to represent the undetermined temporal derivative of stress gradients in (26) and (29). Specifically, we set $\hat{A}^{+'}_l(t) = \hat{A}'_l(t)$ for $t \leq t_{nuc}$ and $\hat{A}^{+'}_l(t) = \hat{A}'_l(t) u(t - t_{nuc})$ for $t > t_{nuc}$. For bricks in

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the VN phase, omitting $\hat{A}_l^{+'}$ in (26) and (29) results in an approximation as the brick solution

$$\sigma_{M,j}^n(x,t) = \begin{cases} \sum_{i=0}^{+\infty}\left\{\sum_{k=0}^{N}\left[\hat{A}_k(0)g_{i,k}^n(x,t)\right] + \sum_{\substack{k=0,\\k\neq l}}^{N}\left[\int_0^t \hat{A}_k'(\tau)g_{i,k}^n(x,t-\tau)d\tau\right]\right\}, & t \le t_{nuc} \\ \sum_{i=0}^{+\infty}\left\{\sum_{k=0}^{N}\left[\hat{A}_k^-(0)g_{i,k}^n(x,t) + \int_0^t \hat{A}_k^{-'}(\tau)g_{i,k}^n(x,t-\tau)d\tau\right]\right. \\ \left. \quad + \sum_{\substack{k=0,\\k\neq l}}^{N}\left[\int_{t_{nuc}}^t \hat{A}_k'(\tau)g_{i,k}^n(x,t-t_{nuc}-\tau)d\tau\right]\right\}, & t > t_{nuc} \end{cases} \quad (30)$$

where N is the number of segments in the brick.

For the VG phase, we substitute $t_0 = t_{nuc}$ into (26) to compute $\sigma_{M,j}^g(x,t)$ under dynamic current density within each segment. The result is then substituted into (18) and (24). By dropping the terms with $\hat{A}_l^{+'}(t)$, an approximation of stress evolution yields

$$\sigma_{M,j}^g(x,t) = \mathcal{B}_{M,j}^g(x,t) + \sum_{i=0}^{+\infty}\left\{\sum_{k=0}^{N}\left[\hat{A}_k(t_{nuc})g_{i,k}^g(x,t-t_{nuc})\right] + \sum_{\substack{k=0,\\k\neq l}}^{N}\int_{t_{nuc}}^t \hat{A}_k'(\tau)g_{i,k}^g(x,t-t_{nuc}-\tau)d\tau\right\}. \quad (31)$$

The brick solutions in (30) and (31) consist of terms that can be precomputed before neural network training and treated as constants, thereby reducing the computational overhead of gradient computation during back-propagation. The terms with $\hat{A}_l^{+'}(t)$ are used to construct the joint solutions.

2) Joint Solution: Fig. 3 presents several partitioning strategies and two detailed scenarios for calculating stress evolution along a cross-shaped interconnect tree. In these scenarios, the partitioning-based approach divides the interconnect tree into two bricks. In the original interconnect tree, the bricks are connected at a joint, which is duplicated to produce the intersection junctions on the adjacent bricks, thereby exhibiting impacts on the stress of both bricks. This influence is controlled by $\hat{A}_l(t)$ and $\hat{A}_l'(t)$, and we further denote the undetermined term $\hat{A}_l^{+'}(t)$ as $\hat{A}_{j,p}^{+'}(t)$ and $\hat{A}_{j,s}^{+'}(t)$ corresponding to the intersection junctions duplicated from the jth joint, on the predecessor and successor bricks, respectively. Subsequently, we introduce a compact multilayer perception (MLP) with RFF embedding $\gamma(\cdot)$ in (12), expressed as $\boldsymbol{y} = f(\gamma(\boldsymbol{B},\boldsymbol{x});\theta)$, where $\boldsymbol{x}$ and θ are the inputs and trainable weights in the neural network $f(\cdot)$. The output $\boldsymbol{y}$ varies from the different locations of joints and time instances, and thus can be rewritten as $y_{j,t}$ to illustrate the approximation of $\hat{A}_{j,p/s}^{+'}(t)$ for the jth joint at time t. Based on (3), we have

$$\hat{A}_{j,p/s}^{+'}(t) = \vec{n}_{j,p/s} \cdot y_{j,t} \quad (32)$$

where $\vec{n}_{j,p} = -1/(W\kappa)$ and $\vec{n}_{j,s} = 1/(W\kappa)$. The impact of the jth joint on stress along each brick during the VN and growth phases, defined as "*joint solution*," is constructed based on (26)

$$\mathcal{J}_{j,p/s}(x,t) = \sum_{i=0}^{+\infty}\int_{t_0}^t \hat{A}_{j,p/s}^{+'}(\tau)g_{i,j,p/s}(x,t-t_0-\tau)d\tau \quad (33)$$

where t_0 is set to 0 for interconnect trees in the VN phase and t_{nuc} in the VG phase. Equation (33) illustrates the locality property of EM-induced stress, where the solution reaches its maximum magnitude at the joint and rapidly decreases along both sides. This phenomenon will be discussed in Section V-A2. As demonstrated in Fig. 3, during the VN phase and along the whole interconnect tree, the stress evolution of the original tree at $t = t_i$ follows $\hat{\sigma}(x,t_i) = \sigma_1(x,t_i) + \mathcal{J}_{1,p}(x,t_i) + \mathcal{J}_{1,s}(x,t_i) + \sigma_2(x,t_i)$ where $\sigma_1(x,t)$ and $\sigma_2(x,t)$ are solutions of brick 1 and brick 2. The spatial coordinates x are first converted to relative coordinates within the bricks, as the inputs for brick and joint solutions. The solutions are then restored to their locations in the original interconnect tree, as illustrated by the stress curves plotted in Fig. 3. Finally, for a general interconnect tree broken into N_b bricks by N_j joints, the stress evolution follows:

$$\hat{\sigma}(x,t) = \sum_{i=1}^{N_b}\sigma_i(x,t) + \sum_{j=1}^{N_j}\left(\mathcal{J}_{j,p}(x,t) + \mathcal{J}_{j,s}(x,t)\right) \quad (34)$$

where $\sigma_i(x,t)$ is the ith brick solution.

3) Compact Solution: According to the decay property of $g(x,t)$, the value diminishes quickly when the traveled distance increases. As a result, we establish a distance threshold $x_{\text{crit}} = 4\sqrt{\kappa t}$, beyond which the value of $g(x,t)$ can be disregarded. Additionally, the convergence of $\mathcal{B}_{M,j}^g(x,t)$ has been proven in [15] and the series $b_k(\cdot)$ and $b_{j,k,M}(\cdot)$ can be truncated following the criteria configured as $M_b = 3$. For the temporal convolution operation in (30), (31), and (33), we employ an eight-point Gauss–Legendre quadrature technology, enhancing computational efficiency while keeping promising accuracy.

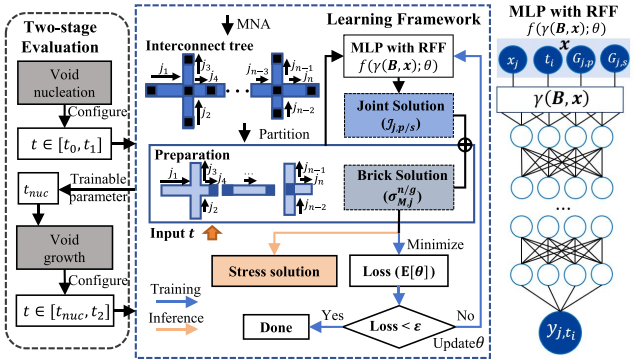


Fig. 4. Framework of the proposed method.

IV. Proposed Learning Framework

The general framework of the proposed method is shown in Fig. 4 consisting of the learning framework and a two-stage evaluation.

A. Learning Framework

1) Preparation: Given an interconnect tree with applied current sources at varying junctions, we first calculate the node voltages and current within each segment using modified nodal analysis (MNA). Based on the brick types outlined in Fig. 2, we partition the interconnect tree into individual bricks and compute the current density within each segment based on the results obtained through MNA.

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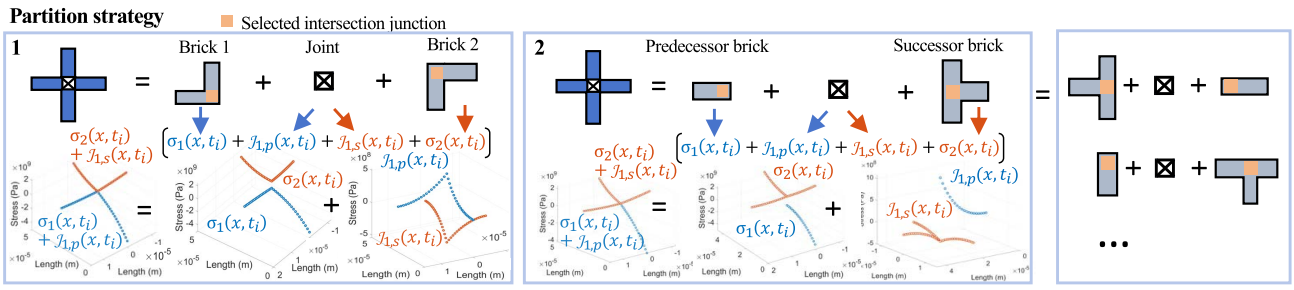


Fig. 3. Partitioning strategy in EM assessment.

The neural network input is structured as a collection $\boldsymbol{x} = \{x_j, G_{j,p}, G_{j,s}, t_i\}$ and the corresponding output y_{j,t_i} is employed to approximate $\hat{A}^{+'}_{j,p/s}$ in (32). Here, we define x_j as the location of the jth joint in the original interconnect tree, and $G_{j,p}$ and $G_{j,s}$ as the average incoming EM driving forces toward the jth joint on the predecessor and successor bricks, respectively. The temporal instances t_i are randomly sampled within the appropriate range using Latin hypercube sampling and logarithmic equidistant sampling strategies as described in [23], with the number of sampling points defined as N_t. The location, EM driving force, time instance, diffusion coefficient and stress are normalized to regular magnitudes following [15], where $x = 10^5 x_o$, $t = 10^{-8} t_o$, $\kappa = 10^{18}\kappa_o$, $G = 10^{-14} G_o$ and $\sigma = 10^{-9}\sigma_o$. The network input is subsequently normalized to the range $[-1, 1]$.

2) Training and Inference: During the training process, following the preparation step, the RFF embedding, as defined in (12), is applied to the inputs before passing them into a compact MLP, as illustrated on the right side of Fig. 4. The matrix $\boldsymbol{B}$ is randomly sampled from a Gaussian distribution and remains fixed throughout the training process. The output y_{j,t_i}, corresponding to different joint locations and N_t time instances, can be regarded as a stress source at the jth joint. The parameters in the MLP with RFF framework are shared when predicting different y_{j,t_i}, enabling efficient mapping from the various joint locations (x_j), time instances (t_i), and EM driving forces $(G_{j,p}, G_{j,s})$ to the corresponding y_{j,t_i}. This output is then propagated outward from the joint and decays over distance. We quantify this propagation as the joint solution, where the resulting $\hat{A}^{+'}_{j,p/s}$ is continuously adjusted by the neural network to build the stress evolution along each brick respecting Korhonen's physical constraints. Substituting the brick solutions and joint solutions into (34), the loss function can be established

$$E[\theta] = \frac{1}{N_t N_j} \sum_{i=1}^{N_t} \sum_{j=1}^{N_j} |\hat{\sigma}_{j,p}(x_j, t_i) - \hat{\sigma}_{j,s}(x_j, t_i)|^2 \qquad (35)$$

where $\hat{\sigma}_{j,p}(x,t)$ and $\hat{\sigma}_{j,s}(x,t)$ are the stress solutions in the predecessor and successor bricks connected with the jth joint located at x_j, respectively. Equation (35) is minimized to penalize the deviations of (4) to approximate joint solutions in (33).

B. Two-Stage EM Evaluation

The proposed two-stage stress evolution in Fig. 4 is divided by the switching of the EM phase. Initially, all the bricks partitioned from the interconnect tree are in the VN phase and the temporal inputs of the learning framework are obtained through random sampling within the time range $[t_0, t_1]$. After convergence of the learning framework, if the stress evolution exceeds the critical level at t_1, the VN time $t_{nuc} \in [t_0, t_1]$ becomes a trainable parameter. The objective is to adjust this parameter so that the maximum stress at the junctions equals the critical level. In this evaluation, voids are assumed to occur at junctions, as no preexisting voids and internal voids are considered in the stress modeling [10]. Once the interconnect tree enters the VG phase, the second stage configures the training time range as $[t_{nuc}, t_2]$. To compute the brick solutions (30) and (31) in this stage, we load the previous learning model to determine the coefficient $\hat{A}^{-'}_{l}$ through outputs at different locations and specific time instances, and $\mathcal{B}^{g}_{M,j}$ through the approximated stress evolution at t_{nuc}. Following a training process similar to that of the first stage, the learning framework can provide the post-voiding stress evolution during the inference phase.

V. Experimental Results

In this section, we present the stress evolution results obtained using the proposed approach during the VN and VG phases. First, we validate the accuracy of the brick solution across various interconnect shapes, and provide a detailed procedure for partitioning an interconnect tree and combining stress solutions along the subcomponents. We then evaluate the training acceleration achieved by incorporating RFF embeddings into a fully connected neural network, compared with utilizing a single MLP as described in [21] and [22]. Finally, we demonstrate the scalability of the proposed approach and the performance enhancements over existing methods. The proposed model and the stress model in [21] are carried out in Python 3.9.5 using PyTorch 1.12 and Tensorflow 1.14, respectively. The competing semi-analytical method in [36] is implemented in MATLAB. The experiments are conducted on a Linux machine equipped with an Intel Core i5-10505 CPU with 16 GB of RAM, and an Nvidia RTX 3090 GPU with a 24 GB buffer. The reference solution is obtained from FEM simulations conducted by the COMSOL Multiphysics software [37].

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In the experiments, the RFF embedding ($m = 30$, $\nu = 3$) is followed by a fully connected MLP with three 60-neuron hidden layers. We utilize the Adam optimizer for neural network training and implement an early stopping mechanism when the loss drops below 5×10^{-6}. Xavier initialization is used to initialize the parameters of the MLP. The activation function is tanh, and the training batch size is configured to $N_t = 1,000$. The values of parameters in the physical model are $k = 1.3806 \times 10^{-23}$ J/K, $e = 1.609 \times 10^{-19}$ C, $Z^* = 10$, $E_a = 1.1$ eV, $B = 1 \times 10^{11}$ Pa, $D_0 = 7.5 \times 10^{-5}$ m^2/s, $\rho = 2.2 \times 10^{-8}$ Ω·m, $\Omega = 8.78 \times 10^{-30}$ m^3, $T = 350$ K, $\sigma_{\text{crit}} = 5 \times 10^8$ Pa. Interconnect trees with the number of offshoots exceeding 10% of the total number of segments are tested in the experiments.

A. Accuracy Analysis

1) Single Brick Solution: Considering a single brick as a complete interconnect tree, we calculate the current distribution based on the configured current/voltage sources. Fig. 5(a) provides the length and current density configuration in the straight 9-segment line, T-shaped, and cross-shaped interconnect trees. Along interconnect trees with varying structures, the comparison of stress solutions obtained using brick solutions and COMSOL simulations are shown in Fig. 5(b)–(d), showing agreement between the stress curves. The stress evaluation encompasses both the VN and VG phases, presenting the stress evolution at 10^8 s from the beginning of each EM phase. With a computation time of less than 0.013 s, the relative errors of stress along a straight nine-segment, T-shaped and cross-shaped interconnect trees are 0.03%, 0.34%, 0.89% during the VN phase, and 0.21%, 0.19%, 0.88% during the VG phase. The stress evolution along the straight nine-segment interconnect tree is divided into two single brick solutions by the void located at the interior junction during the VG phase.

2) Brick-Joint Solution: Given a 14-segment interconnect tree with three offshoots, Fig. 6 lists three strategies for partitioning the interconnect tree into: 1) three straight multisegment bricks; 2) three straight multisegment bricks and one T-shaped brick; and 3) three straight multisegment and one cross-shaped brick. This partitioning is achieved by dividing the interconnect tree into bricks with specific structures.

As illustrated in the first partitioning method, we utilize the proposed brick-joint method to compute the stress evolution at $t = 10^8$ s during the VN phase. To enhance the visualization and facilitate a more straightforward comparison of stress curves derived from varying approaches, we convert the 2-D coordinates of the interconnect trees into a 1-D representation, which has been used in [38]. Specifically, each spatial coordinate (x_i, y_i) is transformed into a single coordinate $x_i + y_i$. Initially, we present the single brick solutions in Fig. 7(a), demonstrating that although Korhonen's diffusion kinetics are observed in the individual stress curves, no constraints are enforced at the intersection junctions. The stress solutions in Fig. 7(b) show compatibility with the flux balance constraint in (3), yet further enhancements

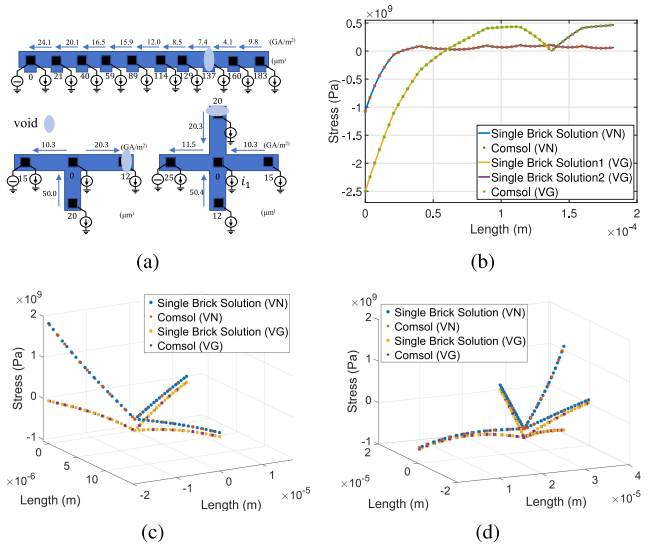


Fig. 5. (a) Configuration of segment lengths and current densities on different single bricks, and the comparison of stress evolution on a (b) straight 9-segment line, c) T-shaped, d) cross-shaped brick at $t = 10^8$ s in the VN and VG phase.

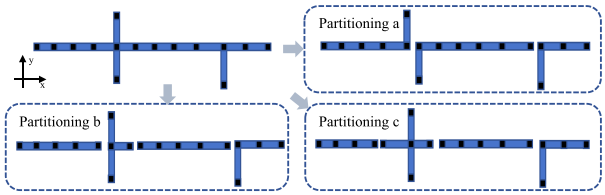


Fig. 6. Three partitioning methods for the 14-segment interconnect tree.

are required to fully satisfy the stress continuity condition in (4). Compared with the single brick solution, the complete brick solution shows progress toward approximating the desired stress curves. Neural network training on joint solutions is then conducted, as plotted in Fig. 7(c), illustrating the influence of the combination operation on stress evolution near the joints. Finally, the prediction obtained by the brick-joint solution is shown in Fig. 7(d), accompanied by the overlapping stress solution calculated from COMSOL.

The performance of the proposed method is reported in Table I where t_{pre}, t_{tra} are the preparation time for dataset generation and brick solution calculation, and the training time, respectively. The notations t_{tot}, t_{inf} correspond to the total time required to achieve a prediction model satisfying $t_{\text{tot}} = t_{\text{pre}} + t_{\text{tra}}$ and the inference time for stress assessment. Errors δ_{rel}, $\delta_{\max}$ are the relative errors between the approximations and results from COMSOL, and the maximum relative errors calculated as $\max |\sigma_{\text{Proposed}} - \sigma_{\text{COMSOL}}| / \max |\sigma_{\text{COMSOL}}|$. The errors are computed based on the stress evolution along the interconnect tree at 10 equidistantly sampled time points within the time range [10^6s, 10^8s]. Three different partitioning methods shown in Fig. 6 are conducted for the performance evaluation, indicating that higher accuracy can be attained by utilizing brick types with a higher junction degree, albeit with

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TABLE I
PERFORMANCE OF THE PROPOSED METHOD UTILIZING PARTITIONING METHODS IN FIG. 6 DURING THE VN AND VG PHASES

Method	Void Nucleation						Void Growth					
	Training			*Inference*	*Error*		*Training*			*Inference*	*Error*	
	t_{pre} (s)	t_{tra} (s)	t_{tot} (s)	t_{inf} (s)	δ_{rel} (%)	δ_{max} (%)	t_{pre} (s)	t_{tra} (s)	t_{tot} (s)	t_{inf} (s)	δ_{rel} (%)	δ_{max} (%)
a	0.61	0.21	0.82	0.009	0.17	0.16	0.18	0.28	0.46	0.016	0.54	0.22
b	0.63	0.41	1.04	0.012	0.18	0.15	0.49	0.44	0.93	0.026	0.34	0.17
c	0.65	0.66	1.31	0.012	0.18	0.12	0.56	0.96	1.52	0.028	0.32	0.16

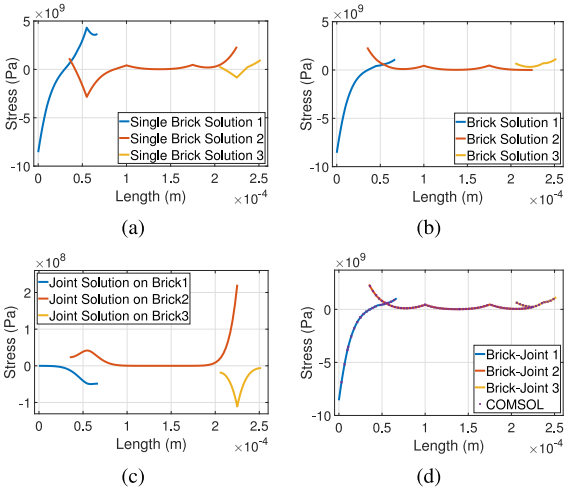


Fig. 7. Stress evolution on a 14-segment interconnect tree at $t = 10^8$ s under partitioning method a in Fig. 6: (a) single brick solutions, (b) brick solutions, (c) joint solutions, (d) summation of (b) and (c) yields the overall brick-joint solution. To enhance the visualization, each 2-D spatial coordinate (x_i, y_i) located in the interconnect tree is transformed into a 1-D coordinate $x_i + y_i$.

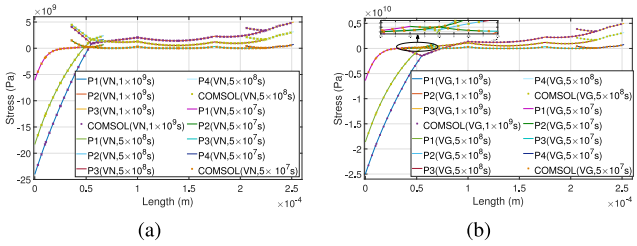


Fig. 8. Comparison of stress evolution on a 14-segment interconnect tree during (a) VN phase, (b) VG phase, where we employ three different partitioning methods shown in Fig. 6. Pi represents the proposed brick-joint solution on the ith brick.

an increase in computation time. The stress prediction at 5×10^7 s, 5×10^8 s, and 10^9 s are plotted in Fig. 8, which are consistent with the results from COMSOL.

3) Physical Parameters and Operating Conditions: In this section, we configure a 40-segment interconnect tree to analyze the impact of varying physical parameters, such as segment width (W) and effective atomic diffusivity (D), as well as operating conditions, including dynamic temperature (T) and current density (j), on EM-induced stress evolution. Fig. 9(a) and (b) depict stress evolution curves at $t = 10^8$ s, obtained after assigning different values of W and D to specific partitions. The widths of the partitions are set to 0.2 μm, 0.1 μm, 0.15 μm, and 0.1 μm, with corresponding effective atomic diffusivity of 9.24×10^{-21} m^2/s, 7.43×10^{-21} m^2/s, 9.28×10^{-21} m^2/s, and 9.90×10^{-21} m^2/s. The relative errors

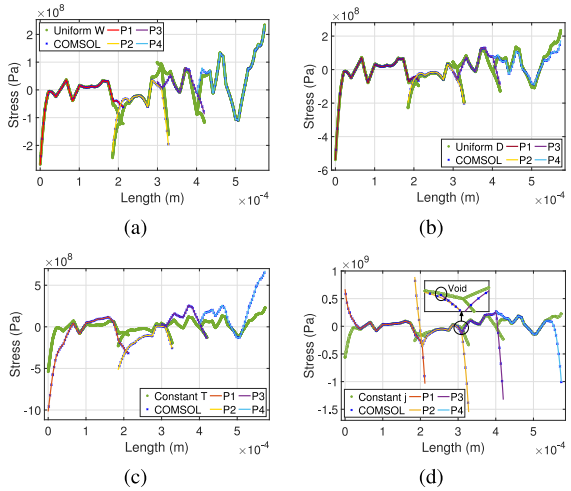


Fig. 9. Comparison of stress evolution on a 40-segment interconnect tree at $t = 10^8$ s considering (a) varying segment width, (b) different effective atomic diffusivity D, and (c) dynamic temperature during the VN phase, and (d) dynamic current density during the VG phase.

of stress results within $t = 10^8$ s, as obtained by the proposed approach, are 0.17% and 0.14%, respectively. Additionally, the stress evolution under constant width ($W = 0.1$ μm) and constant effective atomic diffusivity ($D = 9.24 \times 10^{-21}$ m^2/s) is included in the figures to highlight the significance of incorporating variability in interconnect width and diffusivity driven by stochastic effects in EM assessments.

By setting the dynamic temperature as $T = 350 + 30 \sin(2 \times 10^{-8} \pi t)$ K, the stress evolution at $t = 10^8$ s during the VN phase is plotted in Fig. 9(c), with a relative error of 0.21%. Moreover, the figure includes stress evolution under a constant temperature of 350 K, highlighting the acceleration and deceleration effects of temperature fluctuations on stress build-up. Subsequently, a linear current density is configured for each segment during the VG phase and the EM driving force is set as $G(t) = G_0(1 + 5 \times 10^5 t)$ A/m^2, where G_0 represents the initial EM driving force varying from segments. The stress comparison shown in Fig. 9(d) illustrates the capability of the proposed approach in addressing EM-induced stress considering dynamic current conditions, with a relative error of 0.22%. Stress evolution at 10^8s under constant j is plotted in Fig. 9(d). This highlights the potential for the partitioning-based method to assess EM failure in the VG phase, considering dynamic current densities due to global effects, such as the dynamic IR drop in EM degradation.

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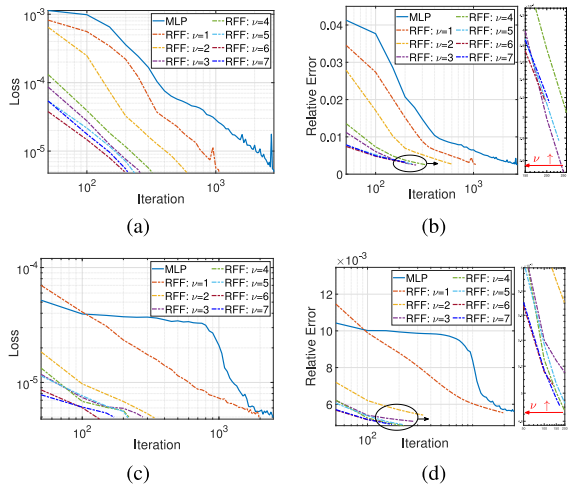


Fig. 10. Loss and relative error versus iterations by employing a conventional MLP [21], [22] and an MLP with RFF embeddings under different ν during (a) and (b) VN phase, and (c) and (d) VG phase. Training is early stopped when the loss reaches 5×10^{-6}.

TABLE II
NUMBER OF TRAINING ITERATIONS/TRAINING TIME (S) TO ACHIEVE THE LOSS OF 5×10^{-6} UNDER DIFFERENT NUMBERS OF HIDDEN LAYERS AND NEURONS PER LAYER IN THE LEARNING FRAMEWORK DURING THE VN PHASE. THE LENGTH OF THE RFF EMBEDDING OUTPUT EQUALS THE NUMBER OF NEURONS PER LAYER

Layers \ Neurons	20	40	60	80
1	2765/7.20	1057/3.54	733/3.03	488/3.94
2	1715/5.21	660/2.78	441/2.42	278/2.86
3	1228/4.34	465/2.35	352/2.73	241/3.02
4	618/5.74	381/4.07	253/3.26	206/3.07

B. Performance Analysis

In this study, an RFF embedding is introduced before passing inputs into a conventional fully connected MLP, resulting in a notable improvement in training speed. Fig. 10 shows the training performance on a 300-segment interconnect tree, reporting the loss and relative error over iterations for a conventional MLP and an MLP with RFF embeddings ($\nu = 1 \sim 7$). Both models are fully connected networks with three hidden layers of 60 neurons each. In the conventional MLP, the weights from the input layer to the first hidden layer are undefined, while in the MLP with RFF embeddings, the mapping follows (12).

The RFF embedding serves as a simple nonlinear transformation for the input, leading to negligible additional training burden. As illustrated in (12), the preferred learning frequency of an RFF-enhanced MLP is controlled by the hyperparameter ν, demonstrated through seven ablation experiments under $\nu = 1 \sim 7$. Fig. 10(a) shows that increasing ν reduces the training iterations required for the loss to reach 5×10^{-6} during the VN phase. While a general improvement in convergence speed is noted as ν increases from 3 to 7, this gain comes at the expense of accuracy in the predicted stress solutions. One potential explanation for the reduction in accuracy is that the neural network tends to interpolate the training data with higher frequency oscillations than those present in the target solution [31]. This trend is also noted during the VG phase.

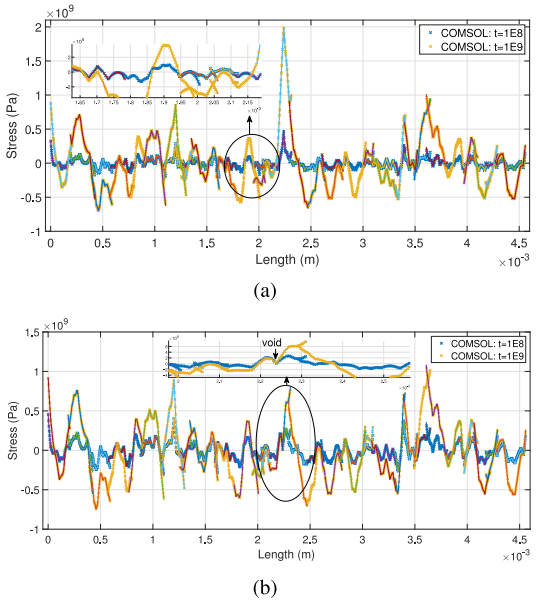


Fig. 11. Stress evolution along a 300-segment interconnect tree at 10^8 s and 10^9 s during (a) VN phase, (b) VG phase. Stress solutions obtained from different bricks are represented in various colors.

To further accelerate convergence as the number of partitions increases, ν can be adjusted to higher values but should be bounded to avoid frequency mismatches between the network and the target solution. In this study, the upper bound for ν is set to 7.

Moreover, we investigate the influence of network structure on training efficiency. Specifically, we set the length of RFF embedding output, defined as 2m in (12), equal to the number of neurons per layer in the cascaded MLP. Table II provides the number of training iterations and the corresponding training time required for achieving a loss of 5×10^{-6} under varying widths and depths of the MLP, and different lengths of the RFF embedding output in the learning framework. The neural network structure should be tailored, considering the tradeoff between the number of iterations and the training time per iteration. In the following experiments, with a sufficient number of neurons in the model, we utilize an RFF embedding configured with $m = 30$, connected to a 3-hidden-layer MLP with 60 neurons per layer.

C. Scalability Analysis

In this section, we evaluate the scalability of the proposed method on interconnect trees with increasing numbers of segments. Fig. 11 illustrates the stress evolution along a 300-segment interconnect tree at $t = 10^8$ s and $t = 10^9$ s obtained from the proposed model and COMSOL. The brick-joint solutions of different bricks are depicted in various colors. The stress curves at the same time instances are indistinguishably overlapped, indicating promising stress predictions in the proposed model. Additionally, the position of void generation is shown in Fig. 11(b).

Table III presents the performance comparison between the proposed approach and COMSOL for interconnect trees with junctions of degree 2–4. In COMSOL simulations, the Finer

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TABLE III
SCALABILITY PERFORMANCE COMPARISON BETWEEN COMSOL AND THE PROPOSED METHOD ON INCREASING n-SEGMENTED INTERCONNECT TREES DURING THE VN AND THE VG PHASE

n-segment	COMSOL	Void Nucleation						Void Growth					
		Training			Speedup	*Inference*	*Error*	*Training*			Speedup	*Inference*	*Error*
	t_{com} (s)	t_{pre} (s)	t_{tra} (s)	t_{tot} (s)	vs COMSOL	t_{inf} (s)	δ (%)	t_{pre} (s)	t_{tra} (s)	t_{tot} (s)	vs COMSOL	t_{inf} (s)	δ (%)
50	56	0.74	0.88	1.62	35×	0.01	0.54	0.28	1.32	1.60	35×	0.02	0.38
100	113	1.00	0.55	1.55	73×	0.03	0.44	0.75	1.57	2.32	49×	0.04	0.54
200	229	1.15	1.13	2.28	100×	0.09	0.71	0.66	1.08	1.74	132×	0.09	0.80
400	604	2.06	3.29	5.35	113×	0.16	0.43	1.59	4.82	6.41	94×	0.16	1.08
600	754	2.87	3.67	6.54	115×	0.23	0.54	2.24	4.20	6.44	117×	0.25	0.84
800	1027	4.25	15.60	19.85	52×	0.25	0.59	3.83	15.61	19.44	53×	0.33	0.96
900	1263	4.38	15.56	19.94	63×	0.31	0.57	3.88	19.34	23.22	54×	0.35	0.43
1000	1456	4.52	20.95	25.47	57×	0.34	0.60	4.15	28.89	33.04	44×	0.42	0.94

mesh setting is utilized, with memory consumption ranging from 1.2 to 4.0 GB of RAM, and the runtime required to calculate stress solutions in each EM phase is defined as t_{com}. With early stopping, the proposed method takes 33.04 s to train a model approximating stress evolution along a 1000-segment interconnect tree, with relative errors of less than 0.94%, where the scale of interconnect trees that can be solved exceeds those mentioned in [21], [22]. The relative errors of stress solutions within 10^8 s are below 1.08% in all test cases and the total training overhead illustrates speedups ranging from 35× to 132× over COMSOL. The memory usage of the proposed approach is $O(n)$, and the training time grows as $O(\beta N_j)$, where β and N_j denote the number of training iterations and the number of joints. During the inference phase, the proposed approach can be viewed as a closed-form expression for stress solutions and demonstrates satisfactory computation speed, requiring less than 0.42 s to achieve stress distribution at 20 equivalent sampled locations per segment. For stress assessment on interconnect trees with 2000, 4000, and 6000 segments, the proposed method requires 57 s, 198 s, and 485 s to converge to a loss of 5×10^{-6}, demonstrating scalability for large-scale interconnects. In these experiments, ν is gradually increased to 6, and the neural network is expanded to 80 neurons per layer, with an RFF embedding size of $m = 40$.

During the VN phase, the relative errors of the model proposed by [21] for solving stress evolution along the 50, 100, and 200-segment interconnect trees, as depicted in Table III, are 1.61%, 18.82%, and 3.99% after total training times of 77.41 s, 124.44 s, and 400.71 s, respectively. The method in [21] employs an MLP to approximate stress dynamics at all junctions. In the 100-segment interconnect tree test case, the stress distributions at all junctions exhibit the highest variance among the three cases, and thus cannot be accurately captured. Furthermore, in the proposed partitioning-based approach, degree-i ($i = 3, 4$) junctions are more likely to be regarded as joints, which exhibit a narrower stress distribution compared with junctions of $i = 1, 2$. Thereby, in addition to benefiting from the reduction in the size of the training dataset induced by partitioning, the proposed method further enhances its approximation ability by narrowing the distribution of desired outputs, achieved by focusing on stress at degree-i ($i = 3, 4$) junctions. The proposed approach demonstrates a speedup of up to 87× over the model from [21] in the three test cases, with higher accuracy in stress analysis. Furthermore, we compare the proposed approach with the semi-analytical method [36] on 100- and 500-segment interconnect trees. The semi-analytical method requires 1.38 s and 67 s to calculate the EM-induced stress during the VN phase, whereas the proposed approach requires 0 s, 0.27 s, and 0.39 s for training the stress model after dividing the interconnect trees into 1, 2, and 3 partitions in the 100-segment case, and 0 s, 0.28 s, and 0.41 s in the 500-segment case. It should be noted that in cases where partitioning the interconnect tree is unnecessary, the proposed method is reduced to analytical expressions. The corresponding inference times are 0.002 s, 0.006 s, and 0.008 s for the 100-segment case and 0.002 s, 0.007 s, and 0.009 s for the 500-segment case, respectively, increasing with the number of partitions and showing a more significant impact on runtime than the segment scale. The proposed approach achieves runtime reductions in both training and inference phases compared with the semi-analytical method.

VI. CONCLUSION

This work introduces a novel partitioning-based approach to predict stress evolution along interconnect trees during the VN and VG phases, utilizing a lightweight neural network. Initially, we present a partitioning strategy to divide a complex interconnect tree into individual bricks through specific joints and derive analytical solutions for each brick. The learning framework is then employed to fine-tune the solutions along bricks, particularly at the intersection junctions connected with neighboring bricks. By simplifying the physics-based constraints through partitioning, the required amount of training data for the learning task is significantly reduced. Moreover, we incorporate RFFs before passing inputs to a multilayer perceptron to enhance the capacity of the neural network to learn features under customized frequencies. The proposed method can be extended to account for the influence of varying physical parameters and operating conditions on stress evolution during EM phases. Experimental results indicate significant computational savings while maintaining competitive accuracy compared with the existing methods. The proposed approach captures continuous multiorder stress while considering dynamic current density, offering a practical alternative to discretization-dominant numerical methods. Although current density simulation is not included here, the promising characteristics of the proposed approach suggest potential

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expansion for comprehensive EM assessments, including void event prediction related to IR drop.

APPENDIX A
DERIVATION OF STRESS EVOLUTION ALONG T-SHAPED BRICKS

In this section, we derive the analytical solution of stress evolution along the T-shaped interconnect trees during the VN phase. Based on (2), we establish the trial solution for the jth segment in the Laplace domain

$$\sigma_j(x,s) = A_j e^{mx} + B_j e^{-mx} \tag{36}$$

where $m = \sqrt{s/\kappa}$ and $\kappa_1 = \kappa_2 = \kappa_3 = \kappa$. We establish the following BCs for a T-shaped interconnect tree

$$\kappa\left(\frac{\partial\sigma_1(x,s)}{\partial x}|_{x=L_0} + G_1\right) + \kappa\left(\frac{\partial\sigma_2(x,s)}{\partial x}|_{x=L_0} + G_2\right) + \kappa\left(\frac{\partial\sigma_3(x,s)}{\partial x}|_{x=L_0} + G_3\right) = 0 \tag{37}$$

$$\sigma_1(x,s)|_{x=L_0} = \sigma_2(x,s)|_{x=L_0} = \sigma_3(x,s)|_{x=L_0}, t > 0 \tag{38}$$

$$\kappa\left(\frac{\partial\sigma_1(x,s)}{\partial x}|_{x=L_1} + G_1\right) = 0$$

$$\kappa\left(\frac{\partial\sigma_2(x,s)}{\partial x}|_{x=L_2} + G_2\right) = 0$$

$$\kappa\left(\frac{\partial\sigma_3(x,s)}{\partial x}|_{x=L_3} + G_3\right) = 0 \tag{39}$$

according to constraints in (3), (4), and (5), respectively. Substituting (36) into (37)–(39), we then obtain a linear matrix equation $\boldsymbol{E}\cdot\boldsymbol{\alpha} = \boldsymbol{\beta}$, where E is a 6×6 matrix, $\boldsymbol{\alpha} = [A_1, B_1, A_2, B_2, A_3, B_3]^T$ describing unknown coefficients, $\boldsymbol{\beta} = [-G_1, -G_2, -G_3, 0, -G_\Sigma]^T$, and $G_\Sigma = \sum_{i=1}^{3} G_i$. Solving this equation gives A_j and B_j as follows:

$$A_j = \frac{-G_j\sum_q\left\{\boldsymbol{h}_q^* e^{m\boldsymbol{\mathcal{H}}_q^*}\right\} - 2\sum_{\substack{k=1,\\k\neq j}}^{3}\left\{G_k\sum_q H_{q,k}^-\right\} + G_\Sigma\sum_q H_q^-}{3m\left(1-\sum_p \mathbf{K}_p^{1,3} e^{-m\mathbf{L}_p^{1,3}}\right)e^{-m\sum_{i=1}^{3}L_i}}$$

$$B_j = \frac{-G_j\sum_q\left\{\boldsymbol{h}_q^* e^{-m\boldsymbol{\mathcal{H}}_q^*}\right\} - 2\sum_{\substack{k=1,\\k\neq j}}^{3}\left\{G_k\sum_q H_{q,k}^+\right\} + G_\Sigma\sum_q H_q^+}{3m\left(1-\sum_p \mathbf{K}_p^{1,3} e^{-m\mathbf{L}_p^{1,3}}\right)e^{-m\sum_{i=1}^{3}L_i}} \tag{40}$$

where

$$H_{q,k}^- = \boldsymbol{h}_q^{\{j,k\}} e^{m\left(\boldsymbol{\mathcal{H}}_q^{\{j,k\}} - L_j\right)}, H_{q,k}^+ = \boldsymbol{h}_q^{\{j,k\}} e^{m\left(\boldsymbol{\mathcal{H}}_q^{\{j,k\}} + L_j\right)}$$

$$H_q^- = \boldsymbol{h}_q^{\{j\}} e^{m\left(\boldsymbol{\mathcal{H}}_q^{\{j\}} - L_j\right)}, H_q^+ = \boldsymbol{h}_q^{\{j\}} e^{m\left(\boldsymbol{\mathcal{H}}_q^{\{j\}} + L_j\right)} \tag{41}$$

and using Algorithm 1 we determine the vectors $\mathbf{L}^{1,3}$, $\mathbf{K}^{1,3}$, $\boldsymbol{\mathcal{H}}^C$, $\boldsymbol{\mathcal{H}}^*$, $\boldsymbol{h}^C$, and $\boldsymbol{h}^*$, where q and p are indices that traverse the entire vectors. $\boldsymbol{\mathcal{H}}^C \triangleq \boldsymbol{\mathcal{H}}(C)$ and $\boldsymbol{h}^C \triangleq \boldsymbol{h}(C)$ are transformations mapping C, a set of indices of segments, to vectors in (41). In Algorithm 1, $\lfloor x\rfloor$ represents the greatest integer less than or equal to x. The calculation operator $A^{l\times 1} +/\cdot B^{1\times l}$ represents the sum/dot product of two matrices $A^{l\times l}$ and $B^{l\times l}$, where the vectors are converted to matrices of uniform size by duplicating specified dimensions of the input vectors. The reshape transform $\mathcal{F}$ follows $\mathcal{F}(X): X\in\mathbb{R}^{l\times p}\rightarrow X\in\mathbb{R}^{1\times lp}$. The operator $a\|b$ represents the concatenation of a vector $a\in\mathbb{R}^n$ and an element $b\in\mathbb{R}$, resulting in a new vector $c\in\mathbb{R}^{n+1}$.

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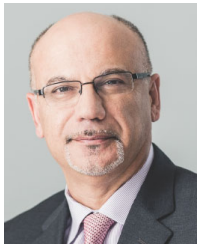
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